

FIT Lecture #7



Fourth Generation: Very High-level Languages

The languages of this generation were considered as very high-level programming languages. The process of developing software using the third generation programming languages required a lot of time and effort that affected the productivity of a programmer.

Fourth Generation: Very High-level Languages

Moreover, most of the third generation programming languages were domain-specific. The fourth generation programming languages were designed and developed to reduce the time, cost and effort needed to develop different types of software applications. Most of the fourth generation programming languages were general-purpose programming languages. This means that most of the fourth generation programming languages could be used to develop software applications related to any domain.

Fourth Generation: Very High-level Languages

During this generation, the concept of Database Management System (DBMS) also evolved tremendously. Therefore, most of the fourth generation programming languages had database related features for working with databases. These languages have simple, English-like syntax rules. Since 4GLs are non-procedural languages, they are easier to use and therefore more user-friendly.

We need to specify WHAT is required rather than specifying How to do it. The following are some of the fourth generation programming languages:

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PowerBuilder, SQL, XBase++, CSS, ColdFusion

Apart from being machine independent, the following are some of the other important advantages of the fourth generation programming languages:

The fourth generation programming languages are easier to learn and use as compared to the third generation programming languages.

These programming languages require less time, cost and effort to develop different types of software applications.

These programming languages allow the efficient use of data by implementing various database concepts.

As compared to the third generation programming languages, these languages required less number of instructions for performing a specific task.

The programs developed in these languages are highly portable as compared to the programs developed in the languages of other generations.

The following are some of the disadvantages of the fourth generation programming languages:

As compared to the programs developed in the programming languages of previous generations, the programs developed in the 4GLs are executed at a slower speed by the CPU.

As compared to the third generation programming languages, the programs developed in these

programming languages require more space in the memory of the computer system.

Fifth Generation: Artificial Intelligence Languages

The programming languages of this generation mainly focus on constraint programming. The constraint programming, which is somewhat similar to declarative programming, is a programming paradigm in which the programmer only needs to specify the solution to be found within the constraints rather than specifying the method or algorithm of finding the desired solution. The major fields in which the fifth generation programming languages are employed are Artificial Intelligence (AI) and Artificial Neural Network (ANN). AI is the branch of computer science in which the computer system is programmed to have human intelligence characteristics.

Fifth Generation: Artificial Intelligence Languages

It helps make computer system so intelligent that it can take decisions on its own while solving various complicated problems. On the other hand, ANN refers to a network that is used to imitate the working of a human brain. ANN is widely used in voice recognition systems, image recognition systems and industrial robotics.

The following are some of the fifth generation programming languages:

Mercury Prolog OPS5

Fifth Generation: Artificial Intelligence Languages

The following are two important advantages of fifth generation programming languages:

The fifth generation languages allow the users to communicate with the computer system in a simple and an easy manner. Programmers can use normal English words while interacting with the computer system.

These languages can be used to query the databases in a fast and efficient manner.



Thanks